Foreign Exchange Risk

Lecture 12

BANK3011
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Objectives of this week's lecture

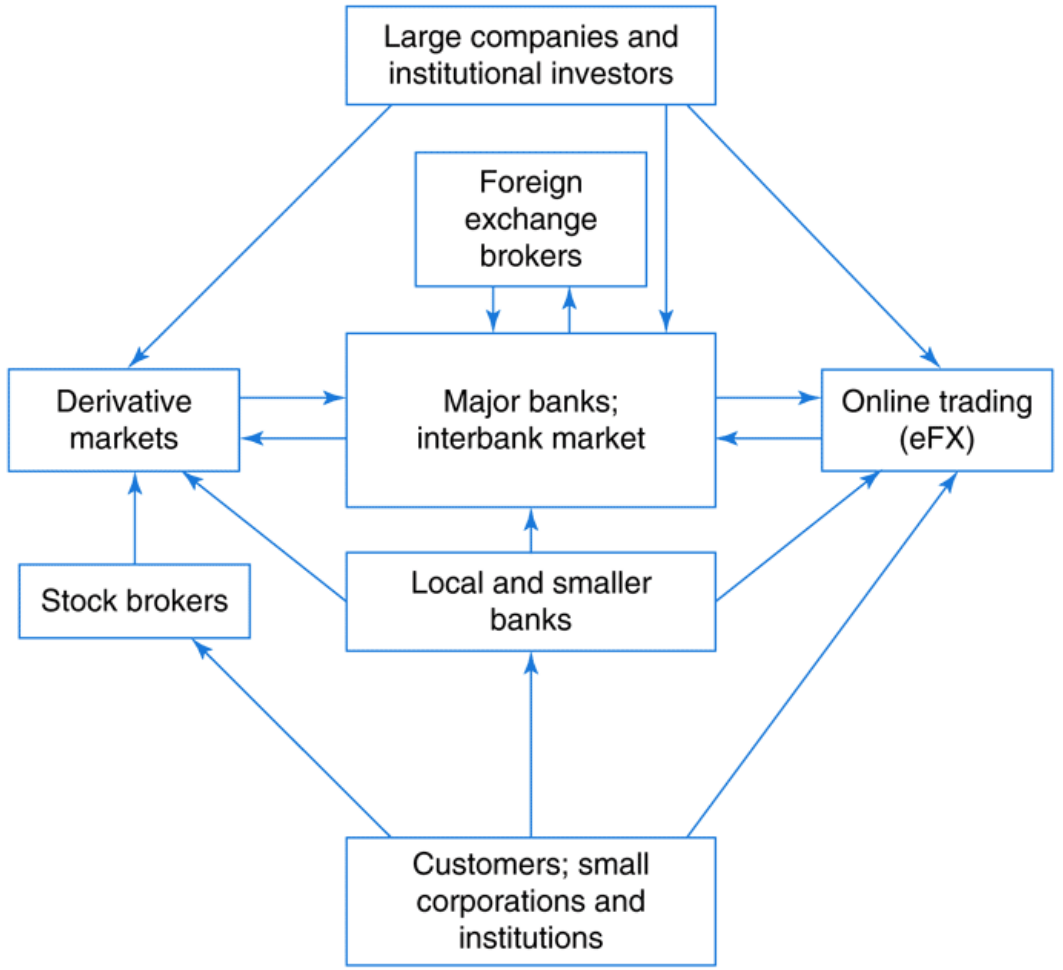
- Foreign exchange rates and transactions.
- Sources of foreign exchange risk exposure.
- Foreign asset and liability positions.
- Risk and hedging
- Interaction of interest rates, inflation, and exchange rates.

1. Foreign Exchange Rate and FX Transactions

- FX market: the global market in which national currencies are bought and sold against one another, and where exchange rates determined.
- As in April 2025 average daily trading volume or turnover in the FX market was estimated at 9.6 trillion USD, 30-35 times greater than daily global GDP!
  - Compared to only $2.55 trillion (daily volume) on the entire US stock market
  - Four days' FX trading volume is comparable to the total market cap of the NYSE (32 trillion dollars)
- The foreign exchange market is an over-the-counter (OTC) market:
  - the FX market consists of a complex international network of informal linkages between key market participants communicating with each other via telephone, fax, e-mail etc.

1. Foreign Exchange Rate and FX Transactions

- The role of banks in the FX market
  - ➢ FX dealers and market makers
    - ✓ ~95% of foreign exchange transactions take place in ***Interbank market*** – wholesale market in which major banks trade with one another.
  - ➢ Take positions; engage in FX arbitrage & speculation.



Note: Our own design, inspired by Figure 1 in Gallaugher and Melville (2004).

1. Foreign Exchange Rate and FX Transactions

FX rate: price at which one currency can be exchanged for another.

- **Direct quote**: U.S. dollars (the home currency) received per unit of foreign currency exchanged.
  - Example: US$0.7668 per Canadian dollar (C$). $0.7668/C$
- **Indirect quote**: Foreign currency received per U.S. dollar (home currency) exchanged.
  - Example: C$1.0342 per US$

Note that the terms *direct* and *indirect* depend on where the quote is obtained.

- Direct quotes in Canada would be expressed in terms of C$ per unit of foreign currency.

1. Foreign Exchange Rate and FX Transactions

- **Spot** foreign exchange transaction: where currencies are traded for *immediate* delivery

- **Forward** foreign exchange transactions: where contracts are made to buy or sell currencies for *future* delivery

Example 1: Fancy Foods, a U.S. company imports meat pies from Australian firms. FF has to pay A$1,000,000 in 90 days in return for supplies. The spot rate is currently AUD1.5 per USD and a banks quotes AUD1.48 per USD for 90-day forward delivery. How many USD will Fancy Foods have to pay in 90 days, if

i) FF chooses no hedge and the spot rate in 90 days is AUD1.55 per USD (i.e. 1 USD = 1.55 AUD)?

ii) FF chooses no hedge and the spot rate in 90 days is AUD1.45 per USD (i.e. 1 USD = 1.45 AUD)?

iii) FF chooses to hedge the risk by entering the forward contract?

1. Foreign Exchange Rate and FX Transactions

i) FF chooses no hedge and the spot rate in 90 days is AUD1.55 per USD (i.e. 1 USD = 1.55 AUD)?

USD payable = 1000000/1.55 = USD645,161.29

ii) FF chooses no hedge and the spot rate in 90 days is AUD1.45 per USD (i.e. 1 USD = 1.45 AUD)?

USD payable = 1000000/1.45 = USD689,655.17

iii) FF chooses to hedge the risk by entering the forward contract?

USD payable = 1000000/1.48 = USD675,675.68

Although the USD payable in scenario 3 is not the lowest, FF removes the volatility in its FX liability using the forward contract. The risk is now shifted to the bank.

1. Foreign Exchange Rate and FX Transactions

The sources of FX risk of banks

- Assets and liabilities denominated in foreign currencies
  - making foreign currency loans
  - issuing foreign currency-denominated debt
- The trading of foreign currencies involves
  - purchase and sale of currencies to complete international transactions
  - facilitating positions in foreign real and financial investments
  - accommodating hedging activities
  - speculation
  - Substantial risk arises via open positions (unhedged positions).

2. Foreign Exchange Risk Exposure

An FI's overall FX exposure in any given currency can be measured by the net position exposure, which is measured in local currency as

$$\text{Net exposure}_i = (\text{FX assets}_i - \text{FX liabilities}_i) + (\text{FX bought}_i - \text{FX sold}_i)$$

- Could match foreign currency assets to liabilities to hedge FX risk.
  - Must also hedge against foreign interest rate risk (by matching durations, for example).
- Financial holding companies have even greater ability to reduce their net exposure to FX risk (diversification through multiple business units such as banking, insurance, fund management, etc.)

2. Foreign Exchange Risk Exposure

- Greater net exposure in foreign currency combined with greater volatility of the foreign exchange rate results in larger potential for losses/gains.
- Dollar loss/gain in currency *i* =

[Net exposure in foreign currency *i* measured in U.S. $] × Shock (Volatility) to the $/Foreign currency *i* exchange rate.

2. Foreign Exchange Risk Exposure

Example 2: The FI has a €2.0 million long trading position in spot euros at the close of business on a particular day. The exchange rate is €0.80/$1, or $1.25/€, at the daily close. Looking back at the daily changes in the exchange rate of the euro to dollars for the past year, the FI finds that the volatility or standard deviation (σ) of the spot exchange rate was 50 basis points (bp). What is the DEAR for this position? Use adverse rate changes in the 95th percentile.

$$\text{DEAR} = \text{Dollar value of position} \times \text{FX volatility}$$

Dollar value of position = €2.0 x 1.25 $/€ = $2.5 million

FX volatility = 1.65σ = 1.65 x 0.005 = 0.00825

DEAR = $2,500,000 x 0.00825= $20,625

3. Foreign Assets and Liabilities

- Mismatches between foreign financial asset and liability portfolios.
- Ability to raise funds from internationally diverse sources presents opportunities, as well as risks.
  - ➢ Exploit market imperfections in foreign banking markets.

Carry trade: If U.S. dollar funding costs 3% but Japanese yen funding costs 1%, a bank can borrow yen, swap into dollars, and invest in higher-yielding U.S. assets—profiting from the spread.

- ➢ Could result in higher return on assets or lower funding costs.

3. Foreign Assets and Liabilities

- Returns are affected by:
  - Spread between costs and revenues.
  - Changes in FX rates can dramatically alter returns.
- Changes in FX rates are not under the control of the FI.
  - Similarities with exposure to interest rate risk.

4. Risk and Hedging

- Hedge can be constructed on- or off-balance-sheet.
  - On-balance-sheet hedge involves changes to on-balance-sheet assets and liabilities to protect against FX risk.

For example: to hedge a Euro payable, a bank can borrow dollars, convert dollars into euros and lend euros to create a synthetic position that will results in Euro inflows and dollar outflows with certainty– a money market hedge. (**Non-examinable**)

  - Off-balance-sheet hedge occurs via position in forward or other derivatives.

Example: Sell the expected proceeds of a British pound loan forward, eliminating future spot FX risk.

- Hedging with forwards
  - FI might choose to remain unhedged on the balance sheet are pursue lower cost alternative.
  - Forward exchange rate is the rate agreed to today for a future delivery of a currency.
    - ✓ Allows the FI to remove the future spot exchange rate uncertainty and other uncertainties relating to returns on investments.

5. Determinants of FX rates

- Global financial markets are increasingly interconnected, so are interest rates, inflation, and foreign exchange rates.
- We now explore how inflation in one country affects its foreign currency exchange rates, focusing on **purchasing power parity** (PPP).
- Next, we examine the relationship between domestic/foreign interest rates and spot/forward foreign exchange rates, known as **interest rate parity** (IRP).

5. Determinants of FX rates

Nominal interest rate R is basically the sum of inflation i and real interest rate r:

For two countries, e.g., EU and United States (US), we have:

$$R^{US} = r^{US} + i^{US}; R^{EU} = r^{EU} + i^{EU}$$

Assuming real interest rates are equal across countries: $r^{US} = r^{EU}$, then

- Therefore, differences in nominal interest rates reflects differences in inflation rates across countries.
- FX rates should adjust in response to differences in inflation rates.
- The Purchasing Power Parity (PPP)

$$i_{\text{Domestic}} - i_{\text{Foreign}} = \frac{\Delta S_{\text{Domestic/Foreign}}}{S_{\text{Domestic/Foreign}}}$$

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5. Determinants of FX rates

- Therefore, differences in nominal interest rates reflects differences in inflation rates across countries.
- FX rates should adjust in response to differences in inflation rates.
  - Imagine you have $2 and can buy one can of Coke in the U.S. In Australia, A$3 also buys a can of Coke.
  - If $2 is worth A$3, things are “balanced” (e.g. AUD1.5 per USD)
  - But if Coke’s price in Australia increases to A$4 (inflation in Australia) and exchange rate stays unchanged, your $2 (or $2*1.5 =A$3) won’t buy as many cans of Coke anymore. Australian dollar should depreciate against USD because its purchasing power has declined.
- The Purchasing Power Parity (PPP)

$$i_{\text{Domestic}} - i_{\text{Foreign}} = \frac{\Delta S_{\text{Domestic/Foreign}}}{S_{\text{Domestic/Foreign}}}$$

5. Determinants of FX rates

- Theory of PPP should generally be applied only for long-term analysis.
- It can take several years for parity to occur.
- Law of one price states that if the market is efficient, identical goods in different countries should have the same price.

(Covered) Interest Rate Parity

- Equilibrium condition is that there should be no arbitrage opportunities available through lending and borrowing across currencies, which requires that.
- Difference in interest rates will equal the percentage spread between forward and spot exchange rates.

$$1 + r_{ust}^{D} = \frac{1}{S_t} \times \left[1 + r_{ukt}^{L}\right] \times F_t$$